Unexpected spousal death and economic security in later life

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Abstract

Research on widowhood typically dates treatment at bereavement, although illness, caregiving, and financial preparation may affect outcomes beforehand. I address this timing problem using 2004–2022 data from the Survey of Health, Ageing and Retirement in Europe (SHARE) and end-of-life interviews to identify spousal deaths preceded by limited forewarning. Combining this classification with propensity-score matching and staggered difference-in-differences models, I estimate the effects of unexpected spousal death on survivors' outcomes. Equivalised income falls, expenditure rises, financial distress increases, and equivalised net wealth declines through real assets and long-term savings, with effects larger for widows. The analysis finds limited labour-market re-entry and no buffering from life insurance, while losses are smaller where survivor provision is more generous. Inheritance data show that wealth declines reflect estate transfers, asset liquidation, and housing adjustment. Findings underscore the importance of treatment timing and joint assessment of resource flows and balance-sheet responses.

JEL codes: D14, G51, J14.

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1 Introduction

The death of a spouse is one of the most severe and enduring economic shocks households can experience. It permanently removes a stream of labour or pension income and often forces a reorganisation of assets, housing, and everyday financial management. In ageing societies, where resources are pooled within couples and scope for labour supply adjustment is limited, spousal death provides a stringent test of whether public insurance and private buffers can protect living standards after a major family shock. This question is especially salient in Europe, where women are substantially more likely than men to outlive their partner and, because of persistent gender gaps in lifetime earnings and contributory histories, often enter widowhood with weaker independent pension entitlements and lower asset holdings (Eurostat, 2024; OECD, 2021).

A large literature in economics and social sciences has examined these issues by comparing the economic outcomes of individuals before and after the death of a spouse. These studies generally model spousal death as a discrete economic shock occurring on the date of bereavement. For many households, however, this date does not mark the beginning of the adjustment process. When death follows a prolonged illness, caregiving demands (Keene & Prokos, 2008) and medical expenditure (Howdon & Rice, 2018; McGarry & Schoeni, 2005) may arise months or years beforehand, stressing household income and savings. Couples may also reallocate financial knowledge and responsibilities prior to the event (Coundouris et al., 2023; Hsu, 2016; Hsu & Willis, 2013), while some financial arrangements may be resolved by the dying spouse before death occurs (Carr & Utz, 2001). Consequently, estimates indexed to the date of bereavement may combine two conceptually distinct processes: economic adjustment in anticipation of death and the effect of losing a spouse itself.

This distinction creates an important identification and interpretation problem. If treatment effectively begins before the recorded event date, pre-bereavement outcomes are already affected by the impending death. Pre-event trends are then not simply a diagnostic for the validity of a research design; they may constitute part of the treatment response. At the same time, estimates beginning at bereavement omit earlier losses and may mischaracterise both the timing and magnitude of the economic consequences. Recent evidence shows that both income and wealth can begin to deteriorate before spousal death (Kapelle & Van Winkle, 2024; Streeter, 2020), and that cumulative losses may be larger when death is expected because part of the adjustment occurs before bereavement (Van Winkle & Leopold, 2022). It remains difficult to determine from conventional widowhood designs how households fare when the loss of a spouse occurs without substantial opportunity for prior adjustment.

This paper addresses the issue by studying unexpected spousal deaths. These provide a setting in which the timing of spousal loss is less entangled with illness-related expenditure, caregiving demands, financial preparation, and other behavioural responses preceding bereavement. Focusing on deaths preceded by limited forewarning therefore reduces a specific source of ambiguity in the timing of treatment and provides a more clearly defined starting point for examining post-bereavement adjustment. The analysis contributes not only new evidence on the economic consequences of spousal death, but also a clearer interpretation of what estimates beginning at bereavement capture. This distinction is also important for understanding how households absorb the shock. When the opportunity to prepare is limited, greater weight is placed on adjustment after the event through labour supply, public and private insurance, estate distribution, portfolio decumulation, housing and living arrangements, and support from family or other social networks. Europe provides a useful setting in which to examine these mechanisms because they operate within diverse institutional and economic environments.

To examine these questions, I use longitudinal data from the Survey of Health, Ageing and Retirement in Europe (SHARE) for 2004–2022. SHARE follows older adults across Europe and provides rich and detailed information on socio-demographic characteristics, health, family networks, and household finances. A distinctive feature of SHARE is its end-of-life interview, administered to a proxy respondent, most often the surviving spouse and otherwise close relatives, which records the circumstances surrounding the respondent's death and the final year of life. I exploit this information to identify unexpected spousal deaths, meaning spousal deaths preceded by limited forewarning, for which the date of bereavement more closely approximates the onset of the household's economic adjustment. The approach is related to studies exploiting sudden deaths to obtain plausibly exogenous variation in household resources, including work on unexpected inheritances (Andersen & Nielsen, 2011; Druedahl & Martinello, 2022), but applies this logic to the timing and interpretation of the economic consequences of spousal loss.

I combine this classification with propensity-score matching and staggered difference-in-differences models. Individuals experiencing an unexpected spousal death are compared with continuously partnered individuals who are similar in observed characteristics before bereavement. The event study traces the evolution of outcomes before and after the death and provides evidence on whether pre-event trends are consistent with the parallel-trends assumption.

The study considers several complementary dimensions of economic security: equivalised household income and expenditure, the income-to-expenditure ratio, financial distress, equivalised net wealth, and detailed portfolio components. Examining these outcomes jointly is necessary because no single measure provides an adequate account of households' adjustment. Stable expenditure may conceal vulnerability when consumption is sustained through asset decumulation. Conversely, a decline in measured wealth need not represent an equivalent decline in the surviving spouse's welfare if assets have been transferred to heirs or portfolios reorganised following the death. The paper therefore distinguishes changes in current resources and material hardship from balance-sheet responses that may reflect transfers or active adjustment.

The results show that unexpected spousal death constitutes a large and persistent negative shock to late-life economic security. Equivalised income declines by about 22%, while equivalised expenditure rises by about 5%, generating a substantial deterioration in the income-to-expenditure ratio and an increase in financial distress. Adjustment also occurs through household balance sheets: equivalised net wealth declines by 23% over time, driven by significant contractions in real assets and drawdowns of long-term savings. Importantly, these effects are substantially larger for widows. The mechanisms analysis indicates little systematic labour-market re-entry and no observable average buffering effect of private life insurance. By contrast, income losses and increases in financial distress are significantly smaller in countries characterised by more generous survivor provision schemes. Information on heirs further shows that part of the decline in real wealth reflects transfers outside the surviving household. Substantial reductions nevertheless remain when the surviving spouse retains the estate, pointing to a combination of financial strain, asset liquidation, and housing adjustment.

The paper makes two contributions. Its primary contribution is to the identification of the economic consequences of spousal death and severe health shocks. Previous studies have documented reductions in income and increases in poverty following widowhood (Angel et al., 2007; Bound et al., 1991; Haveman et al., 2003; Hungerford, 2001; Sevak et al., 2003; Weir et al., 2002), whereas existing evidence on wealth dynamics is comparatively limited and mixed, particularly in Europe (Angel et al., 2007; Coile & Milligan, 2009; Goda & Streeter, 2021; Hurd & Wise, 1987; Poterba et al., 2018). These studies typically date treatment at bereavement even when the underlying economic adjustment may have begun earlier. Recent work has started to move beyond this event-based perspective by treating widowhood as a process and documenting changes in economic outcomes before spousal death (Kapelle & Van Winkle, 2024; Streeter, 2020; Van Winkle & Leopold, 2022). This body of work establishes the importance of anticipation for understanding the economic trajectories surrounding spousal death. The present paper builds on this insight but focuses on its econometric implications. When an expected death affects household behaviour and economic outcomes before bereavement, the date of death does not represent a common treatment onset: outcomes used to estimate the pre-treatment path may already reflect the impending loss, while estimates pooling expected and unexpected deaths combine distinct adjustment processes. By using detailed end-of-life information to identify deaths preceded by limited forewarning, the paper focuses on cases in which the date of bereavement more closely corresponds to the onset of the household's economic adjustment. The paper thereby provides a more clearly timed and more readily interpretable estimate of economic changes at and after bereavement, while tracing both their short- and long-run dynamics. More broadly, the approach is relevant to research on severe health and family shocks (Böckerman et al., 2025; Coile & Milligan, 2009; Fadlon & Nielsen, 2021; García-Gómez et al., 2013; Goda & Streeter, 2021; Jolly & Theodoropoulos, 2023; Lee & Kim, 2008; Poterba et al., 2018; Rabaté et al., 2025), for which anticipation and gradual adjustment may create similar identification issues.

Second, the paper provides an integrated analysis of how older households absorb the economic shock of spousal death across labour-supply, institutional, private-insurance, balance-sheet, housing, and family-support margins. Existing evidence shows that fatal health shocks can induce surviving spouses to increase labour supply when income losses are large (Fadlon & Nielsen, 2021), while access to survivor benefits, whose continued relevance is debated in many countries (OECD, 2018a), reduces the need for this form of self-insurance and provides valuable liquidity following bereavement (Coyne et al., 2024; Rabaté & Tréguier, 2024). Cross-country research similarly indicates that stronger old-age security systems and more generous survivor provision mitigate poverty, consumption losses, and financial hardship among widows (Bíró, 2013; Rabaté et al., 2025; Siegenthaler, 1996; Van Winkle et al., 2024). Private life insurance may provide less effective protection, with U.S. evidence finding no significant improvement in survivors' long-run economic well-being from sizeable insurance payouts (Harris & Yelowitz, 2018). Income-based outcomes, however, capture only part of households' adjustment. Wealth can buffer expenditure, but its interpretation depends on which assets change and why. Existing U.S. evidence shows that widowhood is associated with adjustments across housing, vehicles, businesses, other real estate, and liquid assets (Coile & Milligan, 2009), while European research highlights downsizing, residential mobility, and moves closer to relatives and care (Bonnet et al., 2010; Kapelle & Van Winkle, 2024). Measured wealth may also decline because assets are transferred to other heirs rather than because the resources retained by the survivor deteriorate (Nicholas & Baum, 2020). Building on these literatures, the paper jointly examines labour-market re-entry, survivor pensions, private life insurance, income, expenditure, and financial distress; decomposes wealth into financial and real assets; uses inheritance information to help distinguish resource losses from estate transfers; and traces housing, living-arrangement, and formal and informal support responses. This integrated approach helps distinguish financial hardship from self-insurance, active decumulation, estate distribution, and housing adjustment, providing a more complete account of post-bereavement economic security.

The remainder of the paper is organised as follows. Section 2 describes the SHARE data, the end-of-life module, and the identification of unexpected deaths. Section 3 presents the empirical strategy. Section 4 reports and discusses the main results, potential mechanisms, and heterogeneity. Section 5 concludes.

2 Data and measures

2.1 Data

This study draws on seven regular panel waves (1, 2, 4, 5, 6, 8, and 9) of the Survey of Health, Ageing and Retirement in Europe (SHARE), spanning 2004–2022. SHARE is a multidisciplinary cross-country longitudinal survey of individuals aged 50 and older, fielded biennially across European countries (Bergmann et al., 2019; Börsch-Supan et al., 2013). I exclude the retrospective SHARELIFE waves (3 and 7) because they use a different questionnaire structure and are not suited to the longitudinal outcome measurement used here. The analysis relies primarily on the edited and multiply imputed data produced by the SHARE survey team. These releases provide five imputed datasets for key economic and covariate measures. Using multiple imputation involves a trade-off: it increases sample retention and reduces bias due to item non-response, but it requires stronger assumptions and more complex inference (Franzese, 2023). In this setting, multiple imputation is appropriate given the otherwise substantial loss of observations. In addition, variables related to living arrangements are not imputed in SHARE. Since missingness in these measures is low (below 2%), I restrict analyses involving living arrangements to complete cases on those variables to keep the covariate set consistent within those specifications. I also exclude observations with imputed marital status (fewer than 1%) to ensure a stable definition of partnership and treatment status across imputations. Table A.1 reports, for the unexpectedly widowed sample used in the main analysis, wave-specific missingness rates for each imputed variable used.

I construct the panel in R by merging the relevant SHARE modules, and I conduct the main analysis in Stata using the *mi* suite to estimate models within each imputed dataset and pool inference across imputations following Rubin's combination rules (Rubin, 1987).

2.2 Sample

The widowed sample includes individuals observed in a couple, defined as being married or in a registered partnership, who become widowed during the observation window and for whom end-of-life information about the deceased partner is available. I exclude cases of remarriage after spousal death. Although rare (fewer than 2% of cases), remarriage constitutes a distinct family transition that may alter household resources and, for example, affect entitlement to survivor pensions in many countries.[1] The final widowed sample thus consists of 4,581 individuals, observed on average four times in an unbalanced panel. The pool of potential controls consists of 60,789 individuals who remain continuously partnered with the same partner throughout the observation window and are observed at least twice.

A central feature of the study is the identification of unexpected spousal deaths using information from SHARE end-of-life interviews administered to proxy respondents after a participant's death. In this sample, the proxy is most often the surviving partner (92%) or an adult child (6%), with only a negligible share of other respondents, supporting the reliability of reported circumstances around death. The interview reconstructs the final year of life and records the timing and cause of death, as well as whether the deceased had been ill prior to death and, if so, the duration of illness. Table 1 reports death causes by categories of illness duration before death.

The classification is informed by medical definitions of sudden death, which distinguish natural from unnatural deaths and treat a subset of acute natural causes (e.g., myocardial infarction, stroke, cardiac arrest) as sudden even though they are disease-related, while many unnatural deaths (e.g., accidents and violence) are also considered sudden and unexpected to relatives

[1] In most OECD countries survivor benefits cease upon remarriage, while in others they may continue temporarily or be converted into a lump-sum payment (OECD, 2018b).

(e.g., Andersen & Nielsen, 2011; Li et al., 2003). In the context of economic adjustment, however, the relevant concept is not clinical suddenness per se, but the degree of forewarning and hence the scope for anticipatory behavioural and financial responses by the household. This approach is consistent with the bereavement literature, which distinguishes sudden from anticipated deaths according to the time available to recognise and prepare for the loss. Carr et al. (2001), for example, measure forewarning as the interval between the survivor's realisation that the spouse is going to die and the death itself, and review operational definitions ranging from hours or weeks for sudden deaths to one month or more for meaningful forewarning. More generally, the anticipatory grief literature emphasises that the relevant mechanism is the opportunity for psychological and practical preparation (Sweeting & Gilhooly, 1990).

I therefore use the reported duration of pre-death illness as a proxy for the household's potential forewarning. This distinction is especially relevant for economic outcomes because expenditure, asset decumulation, caregiving arrangements, and transfers may begin changing well before an anticipated death. Rather than relying solely on the cause of death, which may classify an acute terminal event as sudden despite substantial preceding morbidity, I classify a spousal death as unexpected when (i) the proxy reports no illness before death, (ii) the reported illness lasted less than one month, or (iii) death resulted from an accident. The one-month cut-off is intended to identify deaths preceded by, at most, a brief period of illness and therefore limited scope for prolonged adjustment (Carr et al., 2001). Excluding accidents, deaths classified as unexpected under this definition are largely attributable to acute natural causes (Table 1). Based on these criteria, 1,119 individuals are classified as unexpectedly widowed and represent the treated group.

Table 1. Share of causes of death by categories of illness duration

	Illness duration					
Cause of death	Not ill	Less than one month	One to six months	Six months to one year	More than one year	Total
Cancer	4.9%	9.9%	44.0%	52.5%	40.8%	34.9%
Heart attack	44.1%	30.4%	8.6%	6.7%	9.1%	13.7%
Stroke	10.8%	17.5%	10.2%	8.2%	8.8%	10.7%
Other cardiovascular disease	22.6%	15.4%	10.8%	11.2%	13.4%	13.1%
Respiratory disease	0.0%	5.8%	5.0%	3.6%	5.9%	5.2%
Digestive system disease	2.0%	2.3%	4.0%	2.9%	2.8%	2.9%
Severe infectious disease	0.0%	6.7%	6.8%	4.8%	4.2%	5.0%
Accidents	-	-	-	-	-	1.7%
COVID-19 or related	0.0%	3.6%	1.6%	0.8%	0.2%	1.2%
Other	15.7%	8.3%	8.8%	9.5%	14.5%	11.7%
Individuals	102	941	793	526	2,143	4,581

Notes. When the reported cause of death is an accident, proxy respondents are not asked about illness duration. There are 76 such cases.

2.3 Outcomes

2.3.1 Income, expenditure, and wealth

Each wave, SHARE collects detailed household-level information on annual income, expenditure, and wealth, reported by a designated financial respondent on behalf of the household. Total household income includes all annual net sources. Total household expenditure covers major categories such as housing-related spending, food, and medical expenses. Household net wealth is defined as the sum of real and financial assets minus debts. To ensure comparability across countries and over time, all monetary variables are expressed in constant purchasing power parity (PPP) terms and deflated using the Harmonised Index of Consumer Prices, with Germany 2015 as the reference.

To make monetary resources comparable across households of different sizes, I equivalise income, expenditure, and wealth using the OECD square-root scale, which divides each household aggregate by the square root of household size. This adjustment reflects economies of scale in consumption—shared housing costs are the canonical example—so that a two-person household is not assumed to require twice the resources of a one-person household to attain a similar standard of living. Accordingly, estimated effects are interpreted as changes in resources per equivalent adult, not changes in total household aggregates.

Because monetary variables are highly skewed and can take zero or negative values, I apply the inverse hyperbolic sine (IHS) transformation, which accommodates the full support and approximates the logarithm for large values (Bellemare & Wichman, 2020). For interpretability, I convert IHS coefficients into approximate percentage changes using the arcsinh-linear transformation for dummy regressors proposed by Bellemare & Wichman (2020), combining the Halvorsen-Palmquist (1980) approximation with the Kennedy (1981) small-sample bias correction (see footnote for the exact formula).[2]

2.3.2 Financial distress

Financial distress is measured each wave using the standard question on households' ability to make ends meet. I construct a binary indicator equal to one if respondents report 'great difficulty' or 'some difficulty' in making ends meet, and zero otherwise (i.e., 'fairly easily' or 'easily').

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[2] $\frac{\hat{P}}{100} \approx \exp\left(\hat{\beta} - 0.5\widehat{\mathrm{Var}}(\hat{\beta})\right) - 1$

3 Empirical strategy

To improve comparability between unexpectedly widowed and continuously partnered individuals, the empirical strategy combines propensity score matching (PSM) with difference-in-differences (DiD) estimators. Matching is used to construct a control group that is observationally similar at baseline, while DiD exploits within-individual variation over time to net out time-invariant unobserved characteristics and common shocks.

3.1 Matching

I construct the matched control group with PSM (Rosenbaum & Rubin, 1983). This step is particularly relevant given the large imbalance in size between the always-partnered population and individuals who become unexpectedly widowed. Matching is implemented using pre-treatment characteristics measured at the first available observation for each individual. Defining the baseline in this way avoids conditioning on pre-death deterioration or behavioural adjustments that may already be part of the widowhood process. Consistent with recommended practice under multiple imputation, matching is performed separately within each of the five imputed datasets (Leyrat et al., 2019). The propensity score is estimated using a rich set of baseline covariates: country, gender, year of birth, age (respondent and partner), years of education (respondent and partner), job status, area of residence, whether the household moved since the last interview, whether the respondent received help from outside the household in the last 12 months, BMI, number of chronic conditions and mobility limitations, and baseline values of the financial outcomes. Each treated individual is matched to three nearest neighbours with replacement (Van Winkle & Leopold, 2022), imposing a caliper equal to 0.2 standard deviations of the logit of the propensity score (Austin, 2011).

The size of the matched control group varies only modestly across imputations, ranging from 2,935 to 2,977 individuals. Table A.2 reports standardised mean differences (SMDs) by imputation; all SMDs are below 10%, indicating good covariate balance (Stuart et al., 2013). Table 2 summarises baseline characteristics for treated and matched controls. The treated sample is predominantly female (78%), consistent with gender gaps in longevity. Mean age at widowhood is 73 for women and 76 for men, and partners are typically 3–4 years older. Nearly all respondents are married at baseline (99%), with the remainder in registered partnerships. Around 61% are retired and 15% are still working, reflecting the older age profile of the sample.

Table 2. Summary statistics for treated individuals and controls at their first observation

Variables	Matched control mean	Treated mean
Year of birth	1941.7070	1941.0402
Age	67.5715	68.0742
Partner’s age	70.8093	71.5585
Female	0.7600	0.7793
Foreign country of birth	0.1031	0.1001
Years of education	9.5989	9.3985
Partner’s years of education	10.7120	10.5579
Married	0.9876	0.9911
Current job status		
Retired	0.5987	0.6080
Employed/self-employed	0.1692	0.1546
Unemployed	0.0347	0.0322
Permanently sick or disabled	0.0206	0.0214
Homemaker	0.1681	0.1757
Other	0.0087	0.0080
Self-rated health (US scale)	3.2819	3.2779
Body Mass Index (BMI)	27.3390	27.4284
Number of chronic diseases	1.8592	1.8833
Number of mobility limitations	1.7172	1.7748
Number of Activities of Daily Living (ADL) limitations	0.1965	0.1662
Number of Instrumental Activities of Daily Living (IADL) limitations	0.3514	0.3183
EURO-D	2.5997	2.6057
Self-rated eyesight (reading)	2.9462	2.8817
Self-rated hearing	2.6526	2.5980
Times seen/talked to medical doctor (last 12 months)	6.9430	6.8336
Hospitalised (last 12 months)	0.1317	0.1371
Ever smoked	0.3462	0.3324
Physically inactive	0.1136	0.1099
Self-rated reading skills	2.5004	2.5256
Self-rated writing skills	2.6307	2.6627
Orientation score	3.8388	3.8324
Verbal fluency score	18.4874	18.5142
Numeracy score	3.2210	3.2029
Area of residence		
City	0.2498	0.2511
Town	0.4096	0.4057
Village or rural area	0.3406	0.3432
Household size	2.4145	2.3789
Number of children	2.3495	2.2770
Number of grandchildren	3.1919	3.3709
Number of rooms	3.7734	3.7578
Moved	0.1500	0.1448
Co-living	0.2400	0.2198
Received help	0.2003	0.2122
Net income	10.1353	10.1246
Expenditure	9.2263	9.2238
Income-to-expenditure ratio	1.7825	1.7679
Financial distress	0.4866	0.4920
Net wealth	11.0042	10.9489
Real assets	10.8504	10.8223
Net financial assets	5.5409	5.4701
House value	9.7432	9.7426
Share of house owned	0.7764	0.7787
Vehicles value	5.8281	5.5712
Business value	0.4863	0.4620
Share of business owned	0.0364	0.0349
Other real estate value	2.7636	2.8635
Mortgage	1.2159	1.1679
Gross financial assets	6.8782	6.7560
Bank account	6.0223	5.9232
Savings for long-term	2.0980	1.9150
Bonds, stocks, and mutual funds	1.5944	1.5216
Liabilities	1.3505	1.2949

Notes. Statistics from five multiply imputed datasets; pooled using Rubin’s rule. Robust standard errors. Control and treated means are recovered from regressions of each variable on a treatment indicator. Estimation sample varies across imputations. Treatment group size is 1,119, while control group sizes vary between 2,935 and 2,977. All financial variables, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed.

Table A.3 reports the distribution of unexpectedly widowed individuals by country. Country coverage varies across waves due to SHARE's fieldwork design, with countries entering the panel at different times and treated-sample sizes therefore differing across countries. The main analysis pools countries to estimate an average effect, while additional analyses exploit regional variation to study heterogeneity in adjustment patterns.

3.2 Estimation strategy

The timing of spousal death varies across individuals, yielding a staggered treatment design. Treated individuals are compared with the propensity-score matched sample of individuals who remain continuously partnered throughout the observation window. I begin by estimating the following equation with a two-way fixed effects (TWFE) model that exploits within-individual variation over time:

$$Y_{it} = \beta_0 + \beta_1 DiD_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it} \tag{1}$$

where Y_{it} denotes the outcome for individual i in time (year or wave) t, DiD_{it} is a treatment indicator equal to one in all periods after an individual has experienced spousal death. Under parallel trends and homogeneous treatment effects across treatment cohorts and exposure durations, the coefficient β_1 can be interpreted as the average treatment effect on the treated (ATT). The vector X_{it} includes time-varying covariates such as age categories, labour-market status, health measures, cognitive indicators, risk behaviours, social support, and living arrangements. Because several of these variables may themselves respond to spousal death and constitute potential adjustment mechanisms, I report baseline specifications without time-varying controls and additional specifications that add X_{it}, primarily to improve precision. Individual fixed effects α_i absorb time-invariant heterogeneity (e.g., gender, country of birth, education), while λ_t captures common time shocks. Finally, ε_{it} is the error term. Appendix Tables A.4 and A.5 provide complete definitions of the outcomes and time-varying covariates. Since fieldwork within a wave can span multiple years, baseline specifications include year fixed effects to account for macroeconomic conditions and fieldwork timing that are not perfectly aligned with the wave structure. Specifications replacing year fixed effects with country-by-wave fixed effects are also reported in the Appendix.

Identification therefore relies on the parallel-trends assumption: in the absence of the event, outcomes for treated individuals would have followed the same average trajectory as those of the matched controls. Restricting the analysis to deaths preceded by limited forewarning aligns the recorded event more closely with the onset of household adjustment, reducing the likelihood that pre-treatment outcomes already reflect anticipation.[3] Matching improves balance in observed characteristics, individual fixed effects absorb time-invariant heterogeneity, and the event studies assess differential pre-event trends.

I retain TWFE as the principal regression framework because it can be implemented consistently across the multiply imputed datasets and extended transparently to the interaction specifications used to examine potential moderators and heterogeneous responses. This flexibility is particularly useful for comparing adjustment through employment, survivor provision, private insurance, inheritance, and living arrangements within a common empirical framework. It does not, however, resolve the potential weighting problems generated by staggered treatment timing. I therefore complement the baseline estimates with the group-time ATT estimator of Callaway and Sant'Anna (2021). Because the Stata implementation is not natively integrated with the full multiple-imputation and post-estimation procedure required to pool the complete vector of group-time effects and its covariance matrix, this robustness exercise is implemented using the first imputed dataset. Subsection 4.5 describes this and the other robustness exercises in detail.

To assess pre-trends and the evolution of outcomes after bereavement, I estimate the following event-study specification:

$$Y_{it} = \beta_0 + \sum_{k \neq -1} \beta_k D_{it}^k + X_{it}' \gamma + \alpha_i + \lambda_t + \varepsilon_{it} \tag{2}$$

where D_{it}^k is an indicator equal to one when individual i is observed at event time k, relative to spousal death. The omitted category is $k = -1$, the year preceding spousal death, while $k = 0$ denotes the year in which spousal death occurs. The event window is trimmed to ten years before and after the event, balancing the need to capture long-run adjustment with sufficient precision in each event-time bin. This horizon also allows the analysis to detect anticipatory differences several years before bereavement, as documented by Kapelle and Van Winkle (2024), Streeter (2020), and Van Winkle and Leopold (2022).

Because SHARE interviews are generally biennial and fieldwork can span more than one calendar year, calendar-year event time may yield relatively sparse cells. I therefore report complementary event studies indexed by waves since spousal death in the Appendix. I also present dynamic estimates obtained using the Callaway and Sant'Anna (2021) estimator to verify that the event-study profiles are not driven by the weighting structure of conventional TWFE.

[3] The restriction improves treatment timing but does not imply that spousal death is randomly assigned.

All models are estimated separately in each of the five imputed datasets, and coefficients and variance estimates are combined using Rubin's rules (Rubin, 1987). Standard errors are clustered at the individual level throughout.

4 Results

4.1 Preliminary evidence

Table 3 reports some statistics for unexpectedly widowed individuals, comparing the last interview before widowhood to the first interview after spousal death. These before/after differences reflect the combined influence of spousal death, ageing, and time trends.

Several dimensions shift markedly. Labour-market attachment weakens, with respondents more likely to be retired and less likely to be employed. Health deteriorates on average, as indicated by increases in chronic conditions, functional limitations, and hospitalisations. Depressive symptoms (EURO-D) rise sharply, consistent with a substantial decline in psychological well-being following bereavement, while cognitive measures show more modest changes. Household composition changes sharply, with a (mechanical) reduction in household size, a rise in reported help received from outside the household, and some changes in living arrangements around the transition.

Financial outcomes show a clear pattern of increased strain. Equivalised net income declines, while equivalised expenditure changes little on average, implying a deterioration in the income-to-expenditure ratio. Wealth adjustment is evident both in levels and, more clearly, in composition. Equivalised net wealth falls, driven by declines in the value and ownership of real assets. By contrast, net financial assets do not change significantly, largely reflecting a sharp reduction in liabilities, which offsets declines in long-term savings and in holdings of bonds, stocks, and mutual funds.

Table 3. Statistics before and after spousal death

Variables	Pre	Post	Diff (Post – Pre)
Age	71.587	75.860	4.273***
Current job status			
Retired	0.691	0.737	0.046***
Employed/self-employed	0.109	0.079	-0.030***
Unemployed	0.011	0.004	-0.006*
Permanently sick or disabled	0.017	0.022	0.005
Homemaker	0.159	0.136	-0.023**
Other	0.013	0.021	0.008
Self-rated health (US scale)	3.402	3.529	0.127***
Body Mass Index (BMI)	27.450	26.945	-0.505***
Number of chronic diseases	2.009	2.230	0.221***
Number of mobility limitations	2.051	2.622	0.571***
Number of Activities of Daily Living (ADL) limitations	0.215	0.389	0.174***
Number of Instrumental Activities of Daily Living (IADL) limitations	0.445	0.995	0.550***
EURO-D	2.624	3.912	1.289***
Self-rated eyesight (reading)	2.871	2.925	0.055
Self-rated hearing	2.720	2.875	0.155***
Times seen/talked to medical doctor (last 12 months)	7.582	7.380	-0.202
Hospitalised (last 12 months)	0.144	0.187	0.043***
Ever smoked	0.377	0.371	-0.006
Physically inactive	0.136	0.213	0.077***
Self-rated reading skills	2.526	2.535	0.009
Self-rated writing skills	2.662	2.674	0.012
Orientation score	3.824	3.751	-0.074***
Verbal fluency score	18.137	17.429	-0.708***
Numeracy score	3.228	3.222	-0.006
Area of residence			
City	0.255	0.248	-0.007
Town	0.399	0.417	0.018
Village or rural area	0.346	0.335	-0.011
Household size	2.319	1.316	-1.003***
Number of children	2.245	2.204	-0.041**
Number of grandchildren	3.546	3.567	0.021
Number of rooms	3.734	3.697	-0.037***
Moved	0.123	0.088	-0.036***
Co-living	0.185	0.189	0.004
Received help	0.278	0.421	0.142***
Net income	9.999	9.645	-0.353***
Expenditure	9.175	9.140	-0.034
Income-to-expenditure ratio	1.728	1.511	-0.217***
Financial distress	0.472	0.482	0.010
Net wealth	11.081	10.818	-0.262*
Real assets	10.719	9.854	-0.865***
Net financial assets	6.107	6.389	0.282
House value	9.778	9.155	-0.623***
Share of house owned	0.763	0.685	-0.078***
Vehicles value	5.284	3.329	-1.955***
Business value	0.384	0.191	-0.193***
Share of business owned	0.032	0.019	-0.013***
Other real estate value	2.575	1.861	-0.714***
Mortgage	1.127	0.805	-0.321***
Gross financial assets	7.092	7.028	-0.064
Bank account	6.350	6.530	0.179
Savings for long-term	1.924	1.375	-0.549***
Bonds, stocks, and mutual funds	1.501	1.296	-0.205**
Liabilities	1.057	0.671	-0.386***
Observations	1,119	1,119	2,238

Notes. Statistics from five multiply imputed datasets; pooled using Rubin's rule. Pre/post means are recovered from regressions of each variable on a post-event indicator (standard errors clustered at the individual level); 'Diff' reports the estimated post–pre difference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All financial variables, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed.

4.2 Main results

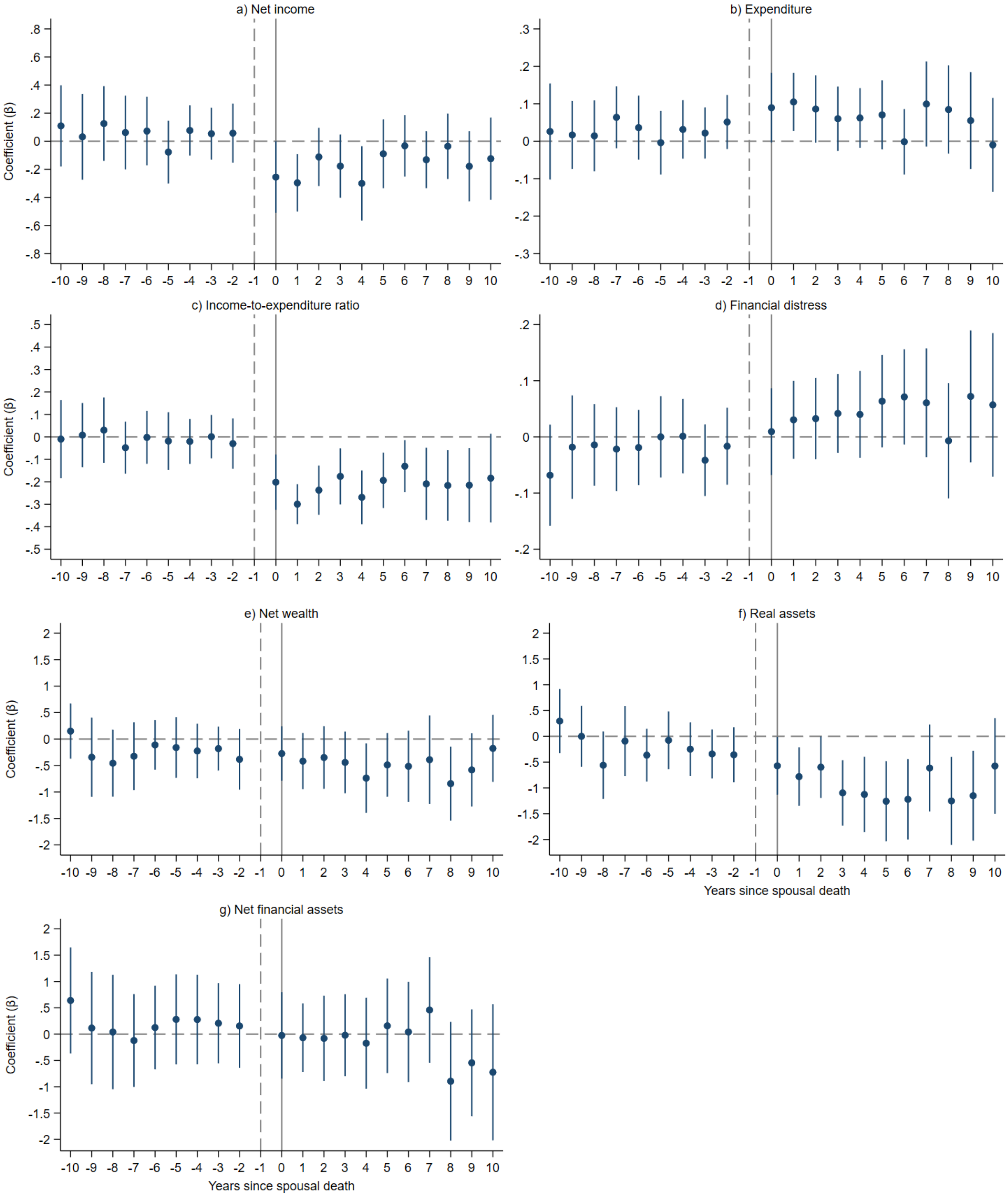
Table 4 presents DiD estimates of the effect of unexpected spousal death on the main outcomes. The results reveal a sizeable 22% reduction in equivalised net income following spousal death ($p < 0.01$), while equivalised expenditure rises by 5% ($p < 0.05$). Together, these effects produce a significant 20% reduction in the income-to-expenditure ratio ($p < 0.01$), signalling increased vulnerability to consumption-smoothing challenges. Consistently, the probability of experiencing financial distress increases by 5.3 percentage points ($p < 0.01$) after bereavement. Wealth responses are also substantial. Equivalised net wealth falls significantly, by roughly 23% ($p < 0.05$). This is driven by a decline in equivalised real assets, whose value drops by 49% ($p < 0.01$), whereas net financial assets do not change significantly. Estimates remain similar when replacing year fixed effects with country-by-wave fixed effects (Table A.6). Disaggregated results in Tables A.7 and A.8 help clarify which portfolio margins move. The decline in real assets is concentrated in housing value ($p < 0.05$) and ownership ($p < 0.01$), vehicles ($p < 0.01$), and other real estate ($p < 0.01$), whereas business ownership and value and outstanding mortgages do not change significantly. On the financial side, gross financial assets and liabilities remain broadly stable. Within financial portfolios, bank account balances and holdings of bonds, stocks, and mutual funds do not change significantly, while long-term savings drop significantly by about 30% ($p < 0.05$). Since this category includes retirement accounts, contractual savings, and whole life-insurance holdings, the pattern is consistent with post-bereavement payouts of life-insurance policies and drawdowns of other illiquid, contract-based savings to generate liquidity.

Table 4. DiD estimates on main outcomes

	Net income		Expenditure		Income-to-expenditure ratio		Financial distress		Net wealth		Net wealth: Real assets		Net wealth: Net financial assets	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DiD	-0.273***	-0.246***	0.0499***	0.0508***	-0.233***	-0.225***	0.0564***	0.0528***	-0.372***	-0.247*	-0.811***	-0.666***	-0.298*	-0.148
	(0.0501)	(0.0504)	(0.0178)	(0.0186)	(0.0254)	(0.0261)	(0.0159)	(0.0161)	(0.133)	(0.148)	(0.157)	(0.172)	(0.179)	(0.186)
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories*, *current job status*, *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity*, *self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score*, *area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Figure 1. Event study



Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors are clustered at the individual level. 95% confidence intervals. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. $k = -1$ is the reference year (vertical dashed line), $k = 0$ is the year in which the event happens (vertical solid line). All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Figure 1 reports event-study estimates that confirm the DiD results while shedding light on the timing of adjustments. Lead coefficients are generally small and not significantly different from zero, supporting the plausibility of parallel trends for the matched sample. At spousal death, equivalised net income drops sharply in the first post-event years and then moves back towards the pre-widowhood baseline in later years. Expenditure increases immediately, consistent with short-run bereavement-related expenses (e.g., funeral costs) but remains elevated throughout the post-period. As a result, the income-to-expenditure ratio falls abruptly at spousal death and shows no meaningful recovery over the subsequent decade. Financial distress also increases after bereavement, but the rise is delayed rather than immediate. This timing is consistent with distress reflecting a cumulative assessment of financial strain rather than an instantaneous response to the event. Finally, wealth adjusts more gradually, consistent with its stock nature. Net wealth displays a gradual decline, driven by real assets, which contract sharply and persistently in the years following bereavement, with only partial recovery. Net financial assets exhibit no clear trend or effect. Figure A.1 reports analogous dynamics using waves since treatment rather than years.

4.3 Potential mechanisms

4.3.1 Survivor benefits, life-insurance policies, and labour-market re-entry

This subsection examines whether the income, expenditure, and distress responses documented above are moderated by three potential mechanisms: behavioural adjustment through labour supply, public insurance through survivor pensions, and private insurance through life-insurance payouts. Although the baseline income measure in SHARE already includes survivor benefits and any reported insurance receipts, these mechanisms may still shape the magnitude of post-bereavement income changes through cross-country differences in benefit generosity and heterogeneity in private coverage.

I begin with labour supply. Column 1 in Table 5 tests whether surviving partners re-enter work after bereavement by regressing an indicator for being employed on the spousal death indicator. The estimated effect is close to zero and statistically insignificant, providing no evidence of systematic labour-market re-entry in this older sample. I then allow the effect of spousal death on economic outcomes to vary by employment status (columns 2, 5, 8, and 11). While there is no significant heterogeneity for expenditure, the income-to-expenditure ratio, or financial distress, the interaction for income is positive and marginally significant (column 2, $\beta_{DiD*Work}$ = 0.537, $p < 0.1$), suggesting that individuals who are employed after bereavement experience a smaller income decline. This is consistent with limited scope for earnings-based self-insurance overall, but some buffering among the subset who returns to work.

I then turn to public survivor provision. To assess whether post-bereavement outcomes vary with the institutional protection available to surviving spouses, I interact the spousal death indicator with country-level survivor-pension replacement rates drawn from the *OECD Pensions Outlook 2018* (OECD, 2018b) and the report *Social Security Programs Throughout the World: Europe, 2018* (The United States Social Security Administration, 2018), the most recent sources providing comparable cross-country measures of survivor-pension generosity. These are assigned at the country level and held fixed across survey waves. They should therefore be interpreted as a time-invariant cross-sectional proxy for institutional generosity, rather than as the benefit rate applicable to each individual at the time of bereavement. Accordingly, the interaction captures whether post-bereavement outcomes differ between countries characterised by more or less generous statutory survivor provision; it does not exploit within-country reforms or identify the causal effect of a change in the replacement rate. The measure also abstracts from individual variation in eligibility, means testing, taxation, actual benefit receipt, and occupational or private coverage. Although the mean age at widowhood in the sample is 73, and therefore above typical eligibility ages (OECD, 2018a), access to and the value of survivor benefits may still vary across individuals. The results are consequently interpreted as evidence of cross-country institutional heterogeneity consistent with a potential buffering role for survivor provision. Table A.9 lists the countries and corresponding replacement rates.

The estimates in Table 5 indicate pronounced institutional heterogeneity. Equivalised income losses are larger in low-replacement settings (column 3, $\beta_{DiD} = -0.648$, $p < 0.01$) and are significantly attenuated where replacement rates are higher ($\beta_{DiD*Survivor\,pension} = 0.714$, $p < 0.01$). Expenditure does not respond significantly in low-replacement settings (column 6), implying a sizeable reduction in the income-to-expenditure ratio (column 9, $\beta_{DiD} = -0.394$, $p < 0.01$), attenuated in more generous settings ($\beta_{DiD*Survivor\,pension} = 0.301$, $p < 0.1$). Financial distress also rises more strongly where replacement rates are lower (column 12, $\beta_{DiD} = 0.184$, $p < 0.01$) and is mitigated in more generous settings ($\beta_{DiD*Survivor\,pension} = -0.232$, $p < 0.01$). At replacement rates commonly observed across the countries in the sample, the point estimates imply economically meaningful differences in post-bereavement outcomes across institutional settings.

Finally, I examine private insurance. Using end-of-life information on whether the deceased held a life-insurance policy and whether the surviving spouse was the beneficiary, I interact spousal death with a life-insurance beneficiary indicator. In the pooled specification (Table 5, columns 4, 7, 10, and 13), the interaction terms are not statistically significant for income, expenditure, the income-to-expenditure ratio, or financial distress, implying no detectable average buffering effect of life insurance in the full sample. This null result is nonetheless difficult to interpret without accounting for the strong cross-country heterogeneity in insurance penetration and in the institutional role of private provision. Insurance take-up is not uniform: while about 11% of treated individuals are reported to be the beneficiary of a life-insurance policy, the share is much higher in Nordic countries such as Sweden (41%). Take-up is also generally higher in Western than in Southern and Eastern Europe (OECD, 2023).

Table 5. DiD estimates on flows and financial distress, role of survivor pensions, life-insurance policies, and labour-market re-entry

	Work	Net income			Expenditure			Income-to-expenditure ratio			Financial distress		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
DiD	0.000791	-0.246***	-0.648***	-0.242***	0.0504***	-0.0157	0.0572***	-0.225***	-0.394***	-0.227***	0.0529***	0.184***	0.0512***
	(0.00482)	(0.0499)	(0.158)	(0.0545)	(0.0187)	(0.0619)	(0.0199)	(0.0259)	(0.0915)	(0.0279)	(0.0163)	(0.0554)	(0.0171)
Work		-0.392*			-0.0820*			-0.169**			0.0721*		
		(0.230)			(0.0443)			(0.0784)			(0.0407)		
DiD*Work		0.537*			0.132			0.243			-0.131		
		(0.299)			(0.303)			(0.193)			(0.136)		
DiD*Survivor pension			0.714***			0.118			0.301*			-0.232**	
			(0.264)			(0.105)			(0.162)			(0.0955)	
DiD*Life insurance				-0.0349			-0.0545			0.0147			0.0140
				(0.0888)			(0.0523)			(0.0707)			(0.0440)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status* (except columns 1, 2, 5, 8, and 11), *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

To address this heterogeneity, I re-estimate the DiD models on a restricted set of countries with consistent early-wave coverage (waves 1–2, plus Estonia given its sample size), grouping them into four clusters: Northern (Sweden, Denmark), Western (France, Netherlands, Belgium, Austria, Germany, Switzerland), Southern (Spain, Italy, Greece), and Eastern (Czechia, Poland, Estonia). Table 6 reports cluster-specific estimates for labour-market re-entry, net income, and financial distress, using the Northern cluster as the reference. Results in column 1 show no heterogeneity in terms of labour-market re-entry. Columns 2 and 4 show sizeable post-event income losses ($\beta_{DiD} = -0.310$, $p < 0.01$) and increased financial distress ($\beta_{DiD} = 0.0960$, $p < 0.05$) in the Northern baseline, and the interaction terms indicate whether these effects differ elsewhere. The estimates suggest that income declines are broadly similar in magnitude across clusters with some income loss mitigation in Western countries ($\beta_{DiD*Western} = 0.185$, $p < 0.05$) and a muted increase in financial distress in the Eastern cluster ($\beta_{DiD*Eastern} = -0.110$, $p < 0.05$). Columns 3 and 5 then introduce the interaction with life insurance and its cluster-specific differentials. The interaction in column 3 is positive in the Northern cluster, consistent with some cushioning of income losses among beneficiaries, although not significant ($\beta_{DiD*Life\ insurance} = 0.188$, *n.s.*). Moreover, the triple interactions for Western and Southern countries are negative and statistically insignificant, and are even significantly negative for Eastern countries ($\beta_{DiD*Life\ insurance*Eastern} = -0.541$, $p < 0.01$), providing no robust evidence that beneficiaries experience smaller losses outside the Nordic context.

Self-selection may represent a plausible explanation for these results. Households with a life-insurance policy have higher equivalised income before bereavement, by about EUR 10,960 ($p < 0.01$), consistent with evidence that income is a key determinant of life-insurance demand (e.g., Li et al., 2007), and therefore have a greater incentive to insure. Insurance payouts may compensate for larger absolute losses among higher-income households, yielding limited differences in net effects.

Table 6. DiD estimates on labour status, income, and distress, by cluster of countries

	Work	Net income		Financial distress	
	(1)	(2)	(3)	(4)	(5)
DiD	-0.000646	-0.310***	-0.367***	0.0960**	0.0862*
	(0.00920)	(0.0644)	(0.0703)	(0.0394)	(0.0480)
DiD*Western	0.0105	0.185**	0.260***	-0.0348	-0.0223
	(0.0112)	(0.0823)	(0.0919)	(0.0446)	(0.0525)
DiD*Southern	0.00614	-0.0931	-0.0340	-0.0241	-0.0113
	(0.00966)	(0.140)	(0.147)	(0.0463)	(0.0536)
DiD*Eastern	-0.00560	0.0895	0.163**	-0.110**	-0.0976*
	(0.0119)	(0.0734)	(0.0776)	(0.0492)	(0.0564)
DiD*Life insurance			0.188		0.0326
			(0.122)		(0.0822)
DiD*Life insurance*Western			-0.285		-0.0460
			(0.192)		(0.107)
DiD*Life insurance*Southern			-0.275		-0.130
			(0.368)		(0.178)
DiD*Life insurance*Eastern			-0.541***		-0.0891
			(0.205)		(0.161)
Controls	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 12,090 and 12,286. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status* (in columns 2 to 5), *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

4.3.2 Inheritance and wealth transfers

A decline in wealth after spousal death may not necessarily imply financial strain, because part of the reduction may reflect planned transfers triggered by the partner's death. To assess the extent to which observed wealth dynamics are mechanically driven by inheritance, I interact the spousal death indicator with measures of asset transfers outside the surviving household. Specifically, I construct two indicators equal to one when the deceased's house (if any) or broader estate is inherited partly or entirely by other heirs, and zero when assets are inherited entirely by the surviving partner or when the deceased held no such asset. Crucially, the end-of-life interviews identify the actual recipients of these transfers at the individual level. The analysis therefore does not need to infer the allocation of inherited assets from statutory inheritance rules or country-level legal frameworks, although these institutions may shape the realised distribution of the estate. Housing is transferred partly or fully to other heirs in 23% of cases. For the broader estate, the corresponding share rises to around 53%. This implies that in almost half of cases, wealth should not decline purely because assets are diverted to other heirs, making it informative to examine whether wealth reductions persist even when transfers are explicitly accounted for.

Table 7 shows that equivalised net wealth (column 1, $\beta_{DiD*House\ transfer} = -0.544$, $p < 0.05$) and value of real assets (column 3, $\beta_{DiD*House\ transfer} = -0.558$, $p < 0.1$) decline more strongly when housing is partly or fully transferred outside the surviving household. Table A.10 shows that housing transfers predict lower housing values (column 1, $\beta_{DiD*House\ transfer} = -0.587$, $p < 0.1$) and lower ownership shares (column 3, $\beta_{DiD*House\ transfer} = -0.136$, $p < 0.01$), and that overall estate transfers lead to a lower value of other real estate (column 12, $\beta_{DiD*Estate\ transfer} = -0.489$, $p < 0.1$). Importantly, however, sizeable declines in real assets remain even after accounting for estate transfers, (Table 7 column 4, $\beta_{DiD} = -0.424$, $p < 0.05$), suggesting that wealth dynamics after spousal death are not explained by inheritance alone and may also reflect asset liquidation or other housing-related adjustments. For example, the value of vehicles falls sharply even when transfer indicators are included (Table A.10 column 6, $\beta_{DiD} = -1.442$, $p < 0.01$), consistent with active decumulation of assets no longer needed in a single-person household, even when these are inherited entirely. Furthermore, the negative coefficient for mortgage value only among those who inherit the estate entirely (Table A.10 column 14, $\beta_{DiD} = -0.299$, $p < 0.05$) may suggest that liquidity generated is used to repay outstanding mortgages.

On the financial side, Table A.11 confirms that gross financial assets and liabilities remain broadly stable across specifications, while long-term savings decline consistently. This pattern suggests that drawdowns of contractual savings and payouts of insurance policies represent a robust component of adjustment after spousal death, alongside the real asset responses documented above.

Table 7. DiD estimates on wealth, role of inheritance

	Net wealth		Net wealth			
			Real assets		Net financial assets	
	(1)	(2)	(3)	(4)	(5)	(6)
DiD	-0.126	-0.0571	-0.542***	-0.424**	-0.0672	-0.100
	(0.160)	(0.193)	(0.190)	(0.214)	(0.207)	(0.257)
DiD*House transfer	-0.544**		-0.558*		-0.362	
	(0.251)		(0.298)		(0.369)	
DiD*Estate transfer		-0.360		-0.458*		-0.0900
		(0.231)		(0.269)		(0.312)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

4.3.3 Additional mechanisms: living arrangements and social support

The results on wealth indicate that the value of real assets declines even after accounting for estate transfers through inheritance, suggesting that part of post-bereavement wealth adjustment may operate through changes in living arrangements rather than estate mechanics alone. This subsection therefore focuses on housing and support responses around spousal death. I focus on whether the surviving partner (i) moves residence, (ii) changes area type (city, town, village/rural), (iii) downsizes, proxied by the number of rooms, (iv) transitions into co-living, and (v) increases reliance on social support, measured by receiving help from outside the household (personal care, practical help, or assistance with paperwork) and by receipt of financial gifts.[4] Table 8 reports DiD estimates for these adjustment margins, while Figure 2 links them to heterogeneity in post-bereavement financial outcomes.

Table 8 provides little evidence of systematic geographic relocation. The probability of moving increases slightly although not significantly, and there are no meaningful within-person shifts across broad area types: changes in the likelihood of living in a city, town, or village/rural area are small and not statistically significant. By contrast, two housing margins stand out. First, unexpected spousal death is associated with downsizing, as the number of rooms in the household's accommodation declines, although the effect is modest in magnitude ($\beta = -0.0491$, $p < 0.05$). Second, widowhood increases the probability of co-living by 3.6

[4] Financial gifts of amount greater than 250€ in the last 12 months. This information is available in SHARE only from wave 4, thus for a subsample.

percentage points ($\beta = 0.0362$, $p < 0.01$), consistent with a shift towards shared living arrangements following bereavement. Social support also rises: the probability of receiving help from people outside the household increases by 6.4 percentage points ($\beta = 0.0641$, $p < 0.01$). In contrast, there is no evidence of a systematic increase in the probability of receiving financial gifts.

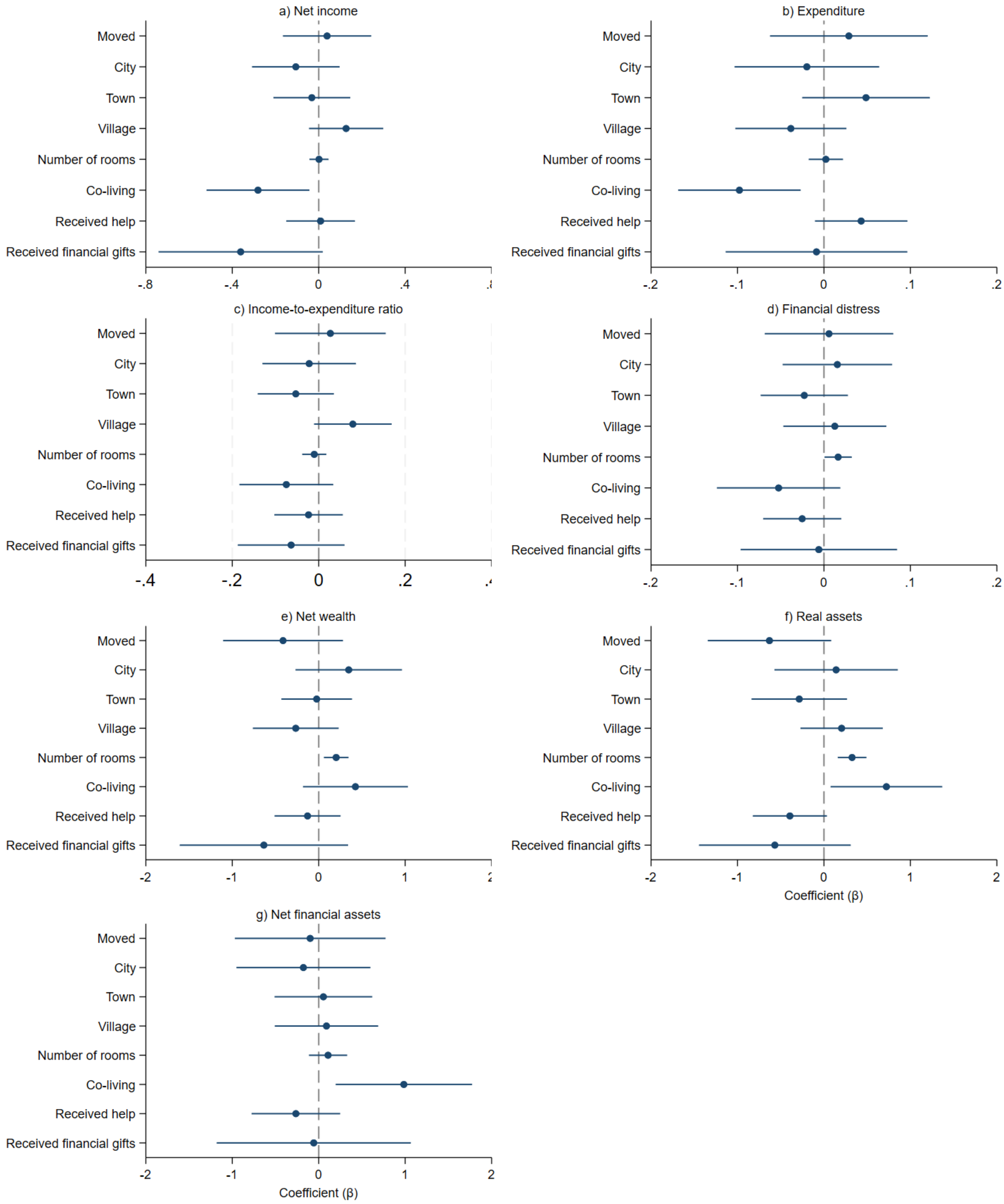
Figure 2 then relates these adjustments to heterogeneity in financial trajectories by plotting the coefficients on interactions between widowhood and each adjustment indicator (with 95% confidence intervals). The patterns suggest that downsizing and co-living are the margins most strongly associated with differential post-bereavement outcomes. Downsizing is associated with larger contractions in net wealth and real assets, while also coinciding with weaker increases in financial distress, plausibly reflecting lower housing-related costs or cost-of-living adjustments. In fact, co-living is associated with lower equivalised income and expenditure (although the income-to-expenditure ratio remains similar), and it is associated with relatively higher post-bereavement values of real and financial assets. Importantly, the interaction between spousal death and receipt of financial gifts is negative for income, indicating that widowed individuals who receive financial transfers exhibit larger income declines than those who do not. This pattern is consistent with financial help responding to stronger economic need. By contrast, moving and changes in area of residence are not systematically related to heterogeneity in financial outcomes, and interactions involving help received are not statistically significant.

Table 8. DiD estimates on living arrangements and social support

	Moved (1)	City (2)	Town (3)	Village (4)	Number of rooms (5)	Co-living (6)	Received help (7)	Received financial gifts (8)
DiD	0.0130 (0.0122)	-0.00945 (0.0103)	0.00908 (0.0132)	0.000370 (0.0100)	-0.0491** (0.0220)	0.0362*** (0.0129)	0.0641*** (0.0185)	0.0115 (0.0124)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168 in columns 1 to 7, and between 10,930 and 11,032 in column 8. Number of groups and clusters varies among imputations. Controls include: *age categories, current job status*, *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity*, *self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy* score, *number of children, number of grandchildren.*

Figure 2. The role of living arrangements and social support



Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors are clustered at the individual level. 95% confidence intervals. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168 except for specifications including *received financial gifts*, for which the sample size varies between 10,930 and 11,032. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status*, *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy* score, *number of children, number of grandchildren.*

4.4 Heterogeneity

The economic consequences of spousal death are unlikely to be uniform. This subsection examines heterogeneity along two salient dimensions. First, I consider gender differences, given women's greater exposure to widowhood and their well-documented disadvantage in later-life economic security. Second, I examine heterogeneity by age at spousal death and by the deceased partner's age at death, since both eligibility conditions and the relevance of adjustment margins may vary over the life cycle: survivor benefits may depend on retirement status or age thresholds, while younger widow(er)s may retain greater scope for labour-market adjustment. Heterogeneity is assessed by augmenting the baseline DiD specification with interactions between the spousal-death indicator and subgroup indicators; detailed estimates are reported in Appendix Tables A.12 and A.13.

The gender patterns are clear and economically meaningful. Among widowers, there is limited evidence of a substantial deterioration in economic security: estimated income losses are small and not significant, financial distress does not rise significantly, and the decline in the income-to-expenditure ratio appears to be driven mainly by higher expenditure rather than by a marked fall in resources. Nor is there a statistically detectable decline in net wealth. By contrast, widows experience a markedly sharper adjustment. Their income and income-to-expenditure ratio decline significantly more than those of widowers, consistent with women entering widowhood with greater dependence on a partner's earnings or pension, and financial distress rises significantly among them. The divergence is even stronger on the wealth margin. For women, net wealth contracts substantially after spousal death, driven primarily by reductions in real assets. Taken together, these results indicate that the economic burden of spousal death falls disproportionately on women, reinforcing the view that widowhood is an important channel through which life-cycle gender inequalities translate into insecurity in old age.

By contrast, heterogeneity by age appears limited in this sample. Splitting the sample at age 60 yields broadly similar average effects across the main outcomes, with no systematic evidence that younger individuals experience different income, distress, or wealth responses than older ones. The same conclusion emerges when stratifying by the deceased partner's age at death: estimated effects are similar whether spousal death occurs at younger or older ages.

4.5 Robustness checks

This subsection evaluates whether baseline results are sensitive to (i) periods of major macroeconomic disruption and fieldwork timing, (ii) potential reporting discontinuities due to changes in the household financial respondent, (iii) violations of the parallel-trends assumption, and (iv) the use of TWFE under staggered treatment timing.

4.5.1 Excluding crisis periods and fieldwork disruptions

Macroeconomic crises can generate atypical dynamics in income, expenditure, and wealth, and may also affect survey fieldwork and reporting. To verify that the main findings are not driven by crisis-specific conditions, I re-estimate the baseline specifications on a restricted sample that excludes both the immediate post-global financial crisis period and the COVID-19 years, retaining interviews conducted between 2010 and 2019. The estimated effects remain similar in sign and magnitude (Table A.14), indicating that the results are not materially driven by crisis episodes and that year fixed effects adequately absorb common macro shocks.

4.5.2 Financial-respondent turnover and reporting discontinuities

In SHARE, household financial information is reported by a single designated household member (the financial respondent). This designation may change across waves, particularly around partner death, raising concerns that estimated changes could partly reflect reporting discontinuities rather than true economic adjustment. I address this by stratifying the sample according to whether the survivor remains the financial respondent after spousal death or whether respondent status changes. Estimated effects are qualitatively similar across both groups (Tables A.15 and A.16), suggesting that results are not driven by respondent turnover.

4.5.3 Placebo event timing

As a further validation exercise, I implement a placebo test by randomly assigning a pseudo-treatment timing to treated individuals in periods prior to their true widowhood transition and re-estimating the baseline event-study specification around this placebo event; no systematic effects should be detected in this case. Figure A.2 confirms this prediction: placebo coefficients are small and generally statistically insignificant around the placebo spousal death event, providing reassurance that the baseline dynamics are not driven by spurious pre-trends or mechanical event-time patterns unrelated to spousal death.

4.5.4 Alternative comparison group: not-yet treated controls

The baseline analysis compares individuals experiencing unexpected spousal death with a matched sample of individuals who remain continuously partnered throughout the observation window. This provides a substantively transparent comparison group and improves balance in observed pre-treatment characteristics. This also avoids using not-yet treated units as controls, which can be particularly problematic when treatment effects are heterogeneous over time and can induce negative weighting in TWFE settings, although the inclusion of never-treated controls does not by itself eliminate the implicit comparisons across treatment cohorts generated by conventional TWFE estimation when treatment timing is staggered. Nevertheless, to assess sensitivity to the construction of the control group, I re-estimate the main specifications using individuals who experience spousal death at a later date as not-yet-treated controls. Results are qualitatively similar (Table A.17), indicating that the main findings are not driven by the particular construction of the matched control group.

4.5.5 Staggered-treatment estimators beyond TWFE

Conventional TWFE estimators may produce difficult-to-interpret averages when treatment timing is staggered and effects vary across treatment cohorts or exposure durations. In particular, already-treated individuals can contribute implicitly to the counterfactual evolution of later-treated cohorts. I address this concern by re-estimating the dynamic effects using the group-time average treatment-effect estimator proposed by Callaway and Sant'Anna (2021). This estimator constructs cohort- and period-specific treatment effects without using already-treated observations as controls and then aggregates them by event time.

Because the Stata implementation is not natively integrated with the multiple-imputation post-estimation procedure needed to pool the complete set of group-time effects and their covariance matrices, I implement this exercise using the first imputed dataset ($m = 1$). Furthermore, given SHARE's predominantly biennial interview structure and the limited number of observations in some calendar-year event-time cells, event time is indexed by waves since spousal death. Figure A.3 reports the resulting dynamic treatment-effect estimates. The profiles are broadly consistent with the baseline TWFE event study, providing support that the principal conclusions are not driven by heterogeneous-treatment weighting in the conventional estimator.

5 Conclusion

Spousal death is usually treated as an economic shock beginning at bereavement. For many households, however, adjustment starts earlier, as illness, caregiving responsibilities, medical expenditure, and the expectation of death affect income, expenditure, saving, and asset allocation. Estimates based on all spousal deaths may therefore combine anticipatory adjustment with the consequences arising at and after bereavement. This paper uses end-of-life information from SHARE to identify deaths preceded by limited forewarning. Combined with propensity-score matching and staggered difference-in-differences models, this classification improves the alignment between the recorded event date and the likely onset of household adjustment, providing a more clearly defined assessment of post-bereavement adjustment. The paper therefore contributes both new evidence on the economic consequences of spousal death and a clearer interpretation of what estimates beginning at bereavement capture.

The results show that losing a spouse is followed by a substantial and persistent deterioration in late-life economic security even when the scope for prolonged prior adjustment is limited. Equivalised income falls by approximately 22%, while equivalised expenditure rises by about 5%, producing a sustained deterioration in the income-to-expenditure ratio and an increase in financial distress. Adjustment also occurs through household balance sheets. Equivalised net wealth declines gradually by approximately 23%, primarily through reductions in real assets and long-term savings, while net financial assets remain broadly stable. These effects are substantially larger for widows, who experience greater income and wealth losses and clearer increases in financial distress. The findings reinforce existing evidence on widowhood-related hardship while showing that a substantial component of the deterioration emerges at and after bereavement itself.

The analysis also illustrates how households adjust across labour-supply, institutional, private-insurance, portfolio, housing, and family-support margins. Labour-market re-entry is limited, consistent with the restricted scope for increasing earnings among an older population. Private life insurance has no significant average moderating effect, whereas income losses and increases in financial distress are smaller in countries characterised by more generous survivor provision. Wealth adjustment reflects several processes. Inheritance information indicates that part of the observed decline in real assets represents transfers outside the surviving household, but substantial reductions remain when the available information suggests that the surviving spouse retains the estate. The accompanying changes in vehicles' value, living arrangements, social support, and long-term savings point to a combination of asset liquidation and active household reorganisation. Considering these responses jointly helps distinguish financial hardship from self-insurance, portfolio adjustment, and intergenerational redistribution.

The findings have implications for policies addressing economic security in later life. Because employment responses are limited, many older survivors have little capacity to replace lost household resources through higher earnings. Protection after bereavement therefore depends on the interaction between public survivor provision, private insurance, accumulated savings and assets, housing, inheritance arrangements, and support from family or social networks. The cross-country evidence is consistent with survivor provision cushioning income losses and financial distress, but the analysis does not identify the optimal generosity or design of survivor-pension systems. More broadly, policies addressing gender inequality in retirement should consider not only inequalities accumulated during working life, but also differences in the resources available to absorb major family shocks after retirement. The larger losses experienced by widows indicate that bereavement can amplify economic disadvantages accumulated over the life course.

Finally, the analysis points to two priorities for future research. First, anticipated and unexpected spousal deaths should be treated as related but distinct empirical settings. The former captures a broader process that may include prolonged caregiving, medical costs, and financial preparation, whereas the latter provides a more clearly timed assessment of adjustment at and after bereavement. Second, the welfare consequences of spousal death cannot be inferred from income, expenditure, distress, or wealth in isolation. Their interpretation requires joint consideration of current resource flows, balance-sheet responses, estate distribution, housing adjustment, and family support. Future research could build on this approach by exploiting reforms to survivor-benefit eligibility and taxation and by examining how means testing, inheritance rules, and housing-market conditions shape household responses. Linking longitudinal surveys to administrative information on pension entitlements, benefit receipt, insurance payments, inheritances, and asset transfers would allow these adjustment margins and their interactions to be measured and identified more directly.

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Appendix

Table A.1. Share of imputed observations (treated), by wave

Variables	Wave 1	2	4	5	6	8	9
Years of education	1.1	1.5	3.2	4.4	4.1	3.5	3.9
Current job status	0.0	0.2	0.3	0.3	0.3	0.2	0.3
Self-rated health (US scale)	0.0	0.0	0.0	0.3	0.1	0.0	0.0
Body Mass Index (BMI)	3.2	3.1	5.1	4.4	4.5	3.4	4.7
Number of chronic diseases	0.0	0.2	0.1	0.0	0.1	0.0	0.1
Number of mobility limitations	0.0	0.0	0.1	0.0	0.1	0.0	0.1
Number of ADL limitations	0.0	0.0	0.1	0.0	0.1	0.2	0.1
Number of IADL limitations	0.0	0.0	0.1	0.0	0.1	0.2	0.1
EURO-D	1.1	2.3	3.0	2.4	4.4	2.9	6.0
Self-rated eyesight (reading)	0.0	0.0	0.1	0.0	0.3	0.2	0.1
Self-rated hearing	0.0	0.0	0.1	0.0	0.2	0.2	0.1
Times seen/talked to medical doctor (last 12 months)	0.0	0.6	1.0	1.6	0.9	1.1	3.7
Hospitalised (last 12 months)	0.0	0.0	0.1	0.1	0.1	0.0	0.6
Ever smoked	0.0	0.0	0.7	0.0	0.0	0.2	0.0
Physically inactive	0.0	0.0	0.1	0.0	0.1	0.0	0.1
Self-rated reading skills	0.0	0.0	0.6	0.3	0.6	0.5	0.9
Self-rated writing skills	0.0	0.0	0.6	0.3	0.6	0.3	0.7
Orientation	0.3	1.5	0.7	0.4	3.7	1.7	3.0
Verbal fluency	0.3	1.7	1.4	3.3	4.3	2.2	3.9
Numeracy	1.4	3.5	2.7	2.3	3.1	2.0	2.7
Number of children	0.0	0.4	0.1	0.5	0.5	0.5	0.4
Number of grandchildren	0.3	1.0	0.4	1.2	2.2	1.5	1.3
Received help	0.0	0.0	0.3	0.3	0.2	0.6	1.1
Received financial gifts	n.a.	n.a.	0.6	0.4	0.6	0.8	1.0
Net income	59.9	56.4	92.3	51.7	45.6	44.6	41.8
Expenditure	37.2	36.3	24.7	39.7	39.9	27.6	22.5
Financial distress	71.1	69.9	92.9	65.3	60.5	52.6	51.8
Net wealth	1.1	0.8	1.0	7.1	12.3	13.0	1.6
Real assets (*hrass*)	60.5	56.2	60.4	61.9	61.2	56.9	57.2
Net financial assets	37.5	36.1	38.6	41.8	44.6	39.0	35.5
House value (*home*)	43.0	39.0	44.7	43.7	38.2	35.6	41.5
Share of house owned (*perho*)	n.a.	n.a.	49.9	52.8	54.5	52.0	51.4
Vehicles value (*car*)	44.7	46.7	22.7	27.1	30.5	31.3	26.5
Business value (*vbus*)	97.7	97.9	47.0	48.8	57.1	64.5	70.5
Share of business owned (*sbus*)	94.0	95.6	97.8	98.5	98.3	98.6	99.3
Other real estate value (*ores*)	81.9	83.4	97.1	96.5	97.4	97.7	98.0
Mortgage (*mort*)	88.0	90.2	85.4	88.3	88.4	89.8	92.2
Gross financial assets	41.5	37.6	91.9	90.5	93.6	95.5	94.9
Bank account	59.3	55.0	44.2	42.7	37.2	35.5	40.7
Savings for long-term	89.7	89.8	50.4	45.2	47.1	49.4	54.4
Bonds, stocks, and mutual funds	88.8	86.7	91.9	90.2	91.4	94.4	96.7
Liabilities	86.2	89.2	89.0	87.9	90.8	91.4	93.0
Observations	349	482	692	735	869	648	701

Notes. Information on financial gifts received and the percentage of the house owned by the household is available from wave 4. For waves 1 and 2, the share of house owned is derived based on other real asset information: *perho = (hrass - (vbus * sbus) - car - ores + mort) / home.*

Table A.2. Standardised mean difference (SMD) between treated and controls after matching

Variables	Imputed dataset				
	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$
Country	0.07	0.10	0.10	0.09	0.09
Year of birth	0.09	0.05	0.06	0.06	0.09
Age	0.08	0.03	0.05	0.05	0.07
Partner's age	0.09	0.06	0.08	0.08	0.09
Female	0.03	0.08	0.05	0.05	0.02
Foreign	0.01	0.02	0.02	0.00	0.00
Years of education	0.05	0.06	0.05	0.03	0.04
Partner's years of education	0.05	0.04	0.03	0.03	0.03
Married	0.05	0.05	0.03	0.03	0.02
Current job status	0.06	0.03	0.04	0.05	0.08
Self-rated health (US scale)	0.00	0.01	0.01	0.01	0.01
Body Mass Index (BMI)	0.00	0.04	0.01	0.03	0.01
Number of chronic diseases	0.03	0.03	0.01	0.01	0.01
Number of mobility limitations	0.02	0.05	0.02	0.00	0.03
Number of ADL limitations	0.03	0.03	0.05	0.07	0.04
Number of IADL limitations	0.03	0.04	0.03	0.05	0.02
EURO-D	0.01	0.02	0.01	0.00	0.02
Self-rated eyesight (reading)	0.05	0.03	0.05	0.08	0.06
Self-rated hearing	0.03	0.04	0.06	0.06	0.07
Times seen/talked to medical doctor (last 12 months)	0.02	0.00	0.02	0.01	0.04
Hospitalised (last 12 months)	0.02	0.02	0.01	0.02	0.03
Ever smoked	0.03	0.06	0.03	0.03	0.01
Physically inactive	0.01	0.03	0.00	0.02	0.01
Self-rated reading skills	0.03	0.03	0.03	0.01	0.01
Self-rated writing skills	0.04	0.03	0.04	0.02	0.01
Orientation	0.03	0.01	0.02	0.01	0.04
Verbal fluency	0.00	0.01	0.01	0.01	0.01
Numeracy	0.03	0.04	0.01	0.00	0.00
Area of residence	0.02	0.02	0.03	0.01	0.01
Household size	0.07	0.04	0.04	0.01	0.04
Number of children	0.05	0.03	0.06	0.06	0.05
Number of grandchildren	0.07	0.06	0.05	0.04	0.06
Number of rooms	0.03	0.04	0.02	0.02	0.01
Moved	0.03	0.01	0.02	0.01	0.02
Co-living	0.06	0.05	0.07	0.01	0.05
Received help	0.01	0.02	0.03	0.03	0.05
Net income	0.02	0.01	0.00	0.01	0.00
Expenditure	0.02	0.02	0.02	0.01	0.02
Income-to-expenditure ratio	0.05	0.02	0.02	0.01	0.00
Financial distress	0.02	0.02	0.01	0.01	0.01
Net wealth	0.01	0.01	0.03	0.01	0.01
Real assets	0.00	0.00	0.03	0.00	0.00
Net financial assets	0.01	0.03	0.01	0.00	0.01
House value	0.00	0.00	0.02	0.01	0.00
Share of house owned	0.01	0.01	0.01	0.01	0.00
Vehicles value	0.07	0.07	0.08	0.05	0.04
Business value	0.00	0.02	0.01	0.01	0.03
Share of business owned	0.00	0.02	0.00	0.01	0.02
Other real estate value	0.03	0.00	0.01	0.03	0.03
Mortgage	0.03	0.00	0.01	0.01	0.02
Gross financial assets	0.04	0.05	0.02	0.01	0.02
Bank account	0.04	0.04	0.01	0.00	0.03
Savings for long-term	0.04	0.06	0.05	0.04	0.04
Bonds, stocks, and mutual funds	0.02	0.02	0.01	0.03	0.01
Liabilities	0.02	0.02	0.00	0.02	0.03
Treated units	1,119	1,119	1,119	1,119	1,119
Control units	2,954	2,970	2,951	2,977	2,935

Table A.3. SHARE fieldwork start wave/year and treated sample size by country

Countries	Fieldwork start wave (year)	Percentage of treated individuals
Austria	1 (2004)	4.1
Germany	1 (2004)	4.0
Sweden	1 (2004)	4.1
Netherlands	1 (2004)	4.2
Spain	1 (2004)	8.0
Italy	1 (2004)	9.4
France	1 (2004/05)	3.8
Denmark	1 (2004)	4.2
Greece	1 (2004/05)	8.7
Switzerland	1 (2004)	3.0
Belgium	1 (2004/05)	4.6
Israel	1 (2005/06)	2.8
Czech Republic	2 (2006/07)	7.7
Poland	2 (2006/07)	5.0
Hungary	4 (2011)	2.9
Portugal	4 (2011)	1.6
Slovenia	4 (2011)	3.2
Estonia	4 (2010/11)	12.8
Luxembourg	5 (2013)	0.5
Croatia	6 (2015)	3.0
Lithuania	7 (2017)	0.5
Bulgaria	7 (2017)	1.0
Finland	7 (2017)	0.1
Latvia	7 (2017)	0.5
Malta	7 (2017)	0.2
Romania	7 (2017)	0.4
Total individuals		1,119 (100%)

Table A.4. List and description of outcome variables

Variable name	Description
Flow variables	
Net income	IHS-transformed equivalised annual net income
Expenditure	IHS-transformed equivalised annual expenditures
Income-to-expenditure ratio	IHS-transformed equivalised income-to-expenditure ratio
Stock variables	
Net wealth	IHS-transformed equivalised net wealth
Real assets	IHS-transformed equivalised value of real assets
Net financial assets	IHS-transformed equivalised value of net financial assets
Financial distress	
Financial distress	Dummy = 1 if household has difficulties in making ends meet; 0 otherwise
Real assets	
House value	IHS-transformed equivalised value of main house
Share of house owned	Share of house owned by the household
Vehicles value	IHS-transformed equivalised value of vehicles
Business value	IHS-transformed equivalised value of business
Share of business owned	Share of business owned by the household
Other real estate value	IHS-transformed equivalised value of other real estate (secondary houses, holiday houses, other real estate, land or forestry)
Mortgage	IHS-transformed equivalised mortgage
Net financial assets	
Gross financial assets	IHS-transformed equivalised value of gross financial assets
Liabilities	IHS-transformed equivalised value of liabilities (excluding mortgages and money owed on land, property or firms)
Gross financial assets	
Bank accounts	IHS-transformed equivalised value of money in bank accounts
Savings for long-term	IHS-transformed equivalised savings for long-term investments (i.e., retirement accounts, contractual savings, whole life-insurance holdings)
Bonds, stocks, and mutual funds	IHS-transformed equivalised value of bonds, stocks, and mutual funds

Table A.5. List and description of time-varying covariates

Variable name	Description
Age	Seven age groups: 50–55, 56–60, 61–65, 66–70, 71–75, 76–80, > 80
Current job status	
Retired	Dummy = 1 if the respondent is retired; 0 otherwise
Employed/self-employed	Dummy = 1 if the respondent is employed or self-employed; 0 otherwise
Unemployed	Dummy = 1 if the respondent is unemployed; 0 otherwise
Permanently sick or disabled	Dummy = 1 if the respondent is unable to work; 0 otherwise
Homemaker	Dummy = 1 if the respondent is homemaker; 0 otherwise
Other	Dummy = 1 if the respondent has other job status; 0 otherwise
Self-rated health (US scale)	Five-point scale (from 1 = excellent to 5 = poor)
Self-rated eyesight (reading)	Five-point scale (from 1 = excellent to 5 = poor)
Self-rated hearing	Five-point scale (from 1 = excellent to 5 = poor)
Body Mass Index (BMI)	
Underweight	Dummy = 1 if BMI < 18.5; 0 otherwise
Healthy weight	Dummy = 1 if 18.5 <= BMI < 25; 0 otherwise
Overweight	Dummy = 1 if 25 <= BMI < 30; 0 otherwise
Obesity	Dummy = 1 if BMI >= 30; 0 otherwise
Number of chronic diseases	Number of diagnosed chronic conditions
Number of mobility limitations	Number of mobility limitations
Number of ADL limitations	Number of limitations in Activities of Daily Living (ADL)
Number of IADL limitations	Number of limitations in Instrumental Activities of Daily Living (IADL)
Times seen/talked to medical doctor	Number of times individual saw or spoke to a doctor in the last 12 months
Hospitalised	Dummy = 1 if individual was hospitalised in the last 12 months; 0 otherwise
EURO-D	EURO depression scale
Ever smoked daily	Dummy = 1 if individual has ever smoked daily; 0 otherwise
Physical inactivity	Dummy = 1 if individual is physically inactive; 0 otherwise
Self-rated reading skills	Five-point scale (from 1 = excellent to 5 = poor)
Self-rated writing skills	Five-point scale (from 1 = excellent to 5 = poor)
Orientation score	Score on the temporal orientation test
Verbal fluency score	Number of animals named in one minute (verbal fluency score)
Numeracy score	Numeracy test score
Area of residence	
City	Dummy = 1 if living in a big city or its suburbs/outskirts; 0 otherwise
Town	Dummy = 1 if living in a large or small town; 0 otherwise
Village or rural area	Dummy = 1 if living in a village or rural area; 0 otherwise
Number of rooms	Number of rooms in the dwelling for household members' personal use (including bedrooms but excluding kitchen, bathrooms, and hallways)
Moved	Dummy = 1 if moved since last interview; 0 otherwise
Co-living	Dummy = 1 if living with someone other than the partner; 0 otherwise
Number of children	Number of children
Number of grandchildren	Number of grandchildren
Received help	Received help from people outside the household in the last 12 months

Table A.6. DiD estimates, including country-by-wave fixed effects

	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Net wealth: Real assets	Net wealth: Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.257***	0.0455**	-0.223***	0.0560***	-0.275*	-0.663***	-0.178
	(0.0495)	(0.0185)	(0.0261)	(0.0156)	(0.149)	(0.172)	(0.183)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories*, *current job status*, *self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity*, *self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score*, *area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.7. DiD estimates on components of real assets

	House		Vehicles value	Business		Other real estate value	Mortgage
	Value	Share		Value	Share		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.473**	-0.0505***	-1.630***	-0.0546	-0.00327	-0.481***	-0.144
	(0.204)	(0.0169)	(0.135)	(0.0901)	(0.00700)	(0.186)	(0.0943)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

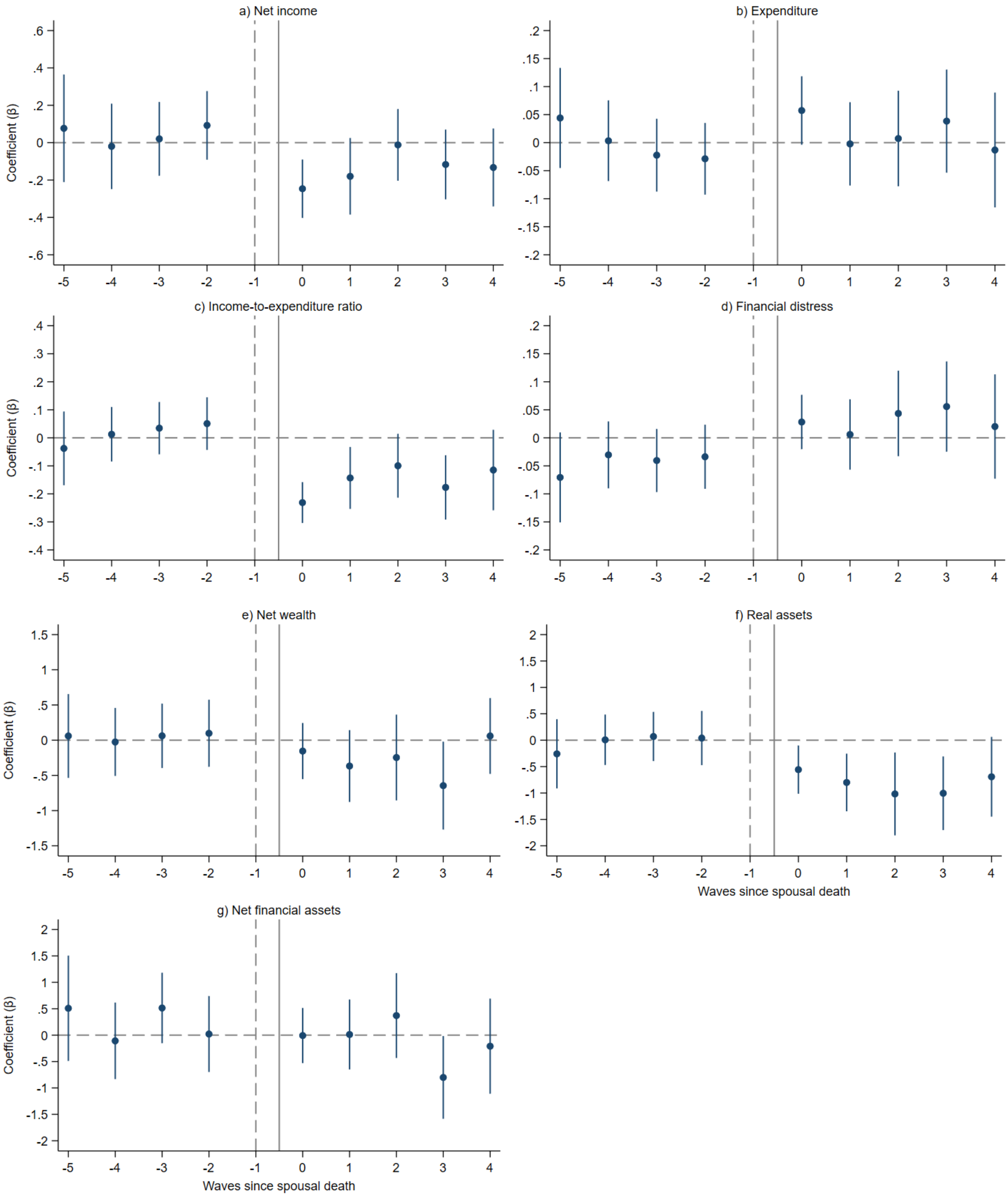
Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.8. DiD estimates on components of net financial assets

	Gross financial assets	Gross financial assets			Liabilities
		Bank account	Savings for long-term	Bonds, stocks, and mutual funds	
	(1)	(2)	(3)	(4)	(5)
DiD	-0.0993	-0.00427	-0.341***	-0.0732	-0.0300
	(0.138)	(0.151)	(0.128)	(0.115)	(0.0933)
Controls	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Figure A.1. Event study, waves since treatment



Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors are clustered at the individual level. 95% confidence intervals. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. $k = -1$ is the reference wave (vertical dashed line), $k = 0$ is the first wave after the event, which happens in between (vertical solid line). All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.9. List of OECD countries and their survivor-pension regimes

	Survivor pension		
Country	Total	Mandatory	Voluntary private
Sweden	19.4%	0%	19.4%
Latvia	26.8%	0%	26.8%
Lithuania	30.4%	0%	30.4%
Greece	37.3%	37.3%	-
Estonia	41.8%	16.9%	24.9%
Bulgaria	50%	50%	-
Finland	50%	50%	-
Romania	50%	50%	-
Denmark	54.9%	0%	54.9%
Germany	55%	55%	-
Malta	55.6%	55.6%	-
France	55.7%	55.7%	-
Czech Republic	57.3%	57.3%	-
Austria	60%	60%	-
Cyprus	60%	60%	-
Hungary	60%	60%	-
Italy	60%	60%	-
Netherlands	60%	0%	60%
Portugal	60%	60%	-
Spain	60%	60%	-
Switzerland	68.4%	68.4%	-
Croatia	70%	70%	-
Slovenia	70%	70%	-
Israel	72.2%	30.6%	41.6%
Luxembourg	79.1%	79.1%	-
Belgium	80%	80%	-
Poland	85%	85%	-

Note. The reported replacement rates refer to survivor-pension schemes measured around 2018. They are used as a time-invariant cross-sectional proxy for institutional generosity and do not capture within-country policy reforms during 2004–2022, individual eligibility conditions, means testing, or variation in occupational and private coverage. *Sources*: OECD (2018b) and The United States Social Security Administration (2018).

Table A.10. DiD estimates on components of real assets, role of inheritance

	House						Business							
	Value		Share		Vehicles value		Value		Share		Other real estate value		Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DiD	-0.342	-0.291	-0.0201	-0.0128	-1.560***	-1.442***	-0.0402	-0.0711	-0.00302	-0.00544	-0.427**	-0.223	-0.156	-0.299**
	(0.218)	(0.231)	(0.0172)	(0.0196)	(0.149)	(0.181)	(0.0961)	(0.109)	(0.00760)	(0.00870)	(0.207)	(0.243)	(0.106)	(0.143)
DiD*House transfer	-0.587*		-0.136***		-0.312		-0.0646		-0.00111		-0.246		0.0519	
	(0.339)		(0.0296)		(0.288)		(0.154)		(0.0121)		(0.369)		(0.177)	
DiD*Estate transfer		-0.345		-0.0713***		-0.355		0.0312		0.00412		-0.489*		0.293*
		(0.297)		(0.0263)		(0.237)		(0.131)		(0.0107)		(0.293)		(0.173)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.11. DiD estimates on components of net financial assets, role of inheritance

	Gross financial assets		Gross financial assets: Bank account		Gross financial assets: Savings for long-term		Gross financial assets: Bonds, stocks, and mutual funds		Liabilities	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
DiD	-0.0772	-0.0430	0.0548	0.0680	-0.461***	-0.332*	-0.123	-0.0581	-0.0939	-0.0656
	(0.150)	(0.173)	(0.165)	(0.181)	(0.142)	(0.176)	(0.129)	(0.159)	(0.103)	(0.133)
DiD*House transfer	-0.0991		-0.265		0.538**		0.223		0.286	
	(0.274)		(0.269)		(0.228)		(0.191)		(0.182)	
DiD*Estate transfer		-0.107		-0.137		-0.0167		-0.0287		0.0673
		(0.211)		(0.210)		(0.209)		(0.186)		(0.159)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.12. DiD estimates on main outcomes, heterogeneity by gender

	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Net wealth: Real assets	Net wealth: Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.113	0.0899**	-0.147***	0.00709	0.271	0.140	0.161
	(0.0922)	(0.0356)	(0.0526)	(0.0331)	(0.235)	(0.231)	(0.329)
DiD*Female	-0.170*	-0.0500	-0.1000*	0.0584*	-0.663**	-1.030***	-0.395
	(0.103)	(0.0405)	(0.0579)	(0.0353)	(0.260)	(0.280)	(0.360)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.13. DiD estimates on main outcomes, heterogeneity by age

											Net wealth			
	Net income		Expenditure		Income-to-expenditure ratio		Financial distress		Net wealth		Real assets		Net financial assets	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DiD	-0.227***	-0.239***	0.0540***	0.0575***	-0.222***	-0.234***	0.0460***	0.0507***	-0.257*	-0.247	-0.729***	-0.707***	-0.180	-0.175
	(0.0506)	(0.0505)	(0.0192)	(0.0193)	(0.0269)	(0.0269)	(0.0171)	(0.0169)	(0.150)	(0.153)	(0.173)	(0.178)	(0.199)	(0.201)
DiD*Age at spousal death > 60	-0.231		-0.0377		-0.0366		0.0791		0.115		0.723		0.367	
	(0.198)		(0.0554)		(0.0905)		(0.0576)		(0.485)		(0.463)		(0.793)	
DiD*Partner's age at death > 60		-0.110		-0.0996		0.139		0.0317		-0.00796		0.596		0.401
		(0.194)		(0.0617)		(0.100)		(0.0606)		(0.521)		(0.500)		(0.912)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 13,988 and 14,168. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.14. DiD estimates, 2010–2019

						Net wealth	
	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Real assets	Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.273***	0.0666**	-0.245***	0.0629**	-0.211	-0.701**	-0.0677
	(0.0760)	(0.0305)	(0.0396)	(0.0264)	(0.266)	(0.313)	(0.299)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 8,200 and 8,278. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.15. DiD estimates on main outcomes, financial respondent status non-switchers

	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Net wealth: Real assets	Net wealth: Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.226***	0.0665***	-0.223***	0.0579***	-0.276	-0.713***	-0.0517
	(0.0614)	(0.0238)	(0.0351)	(0.0188)	(0.185)	(0.207)	(0.223)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

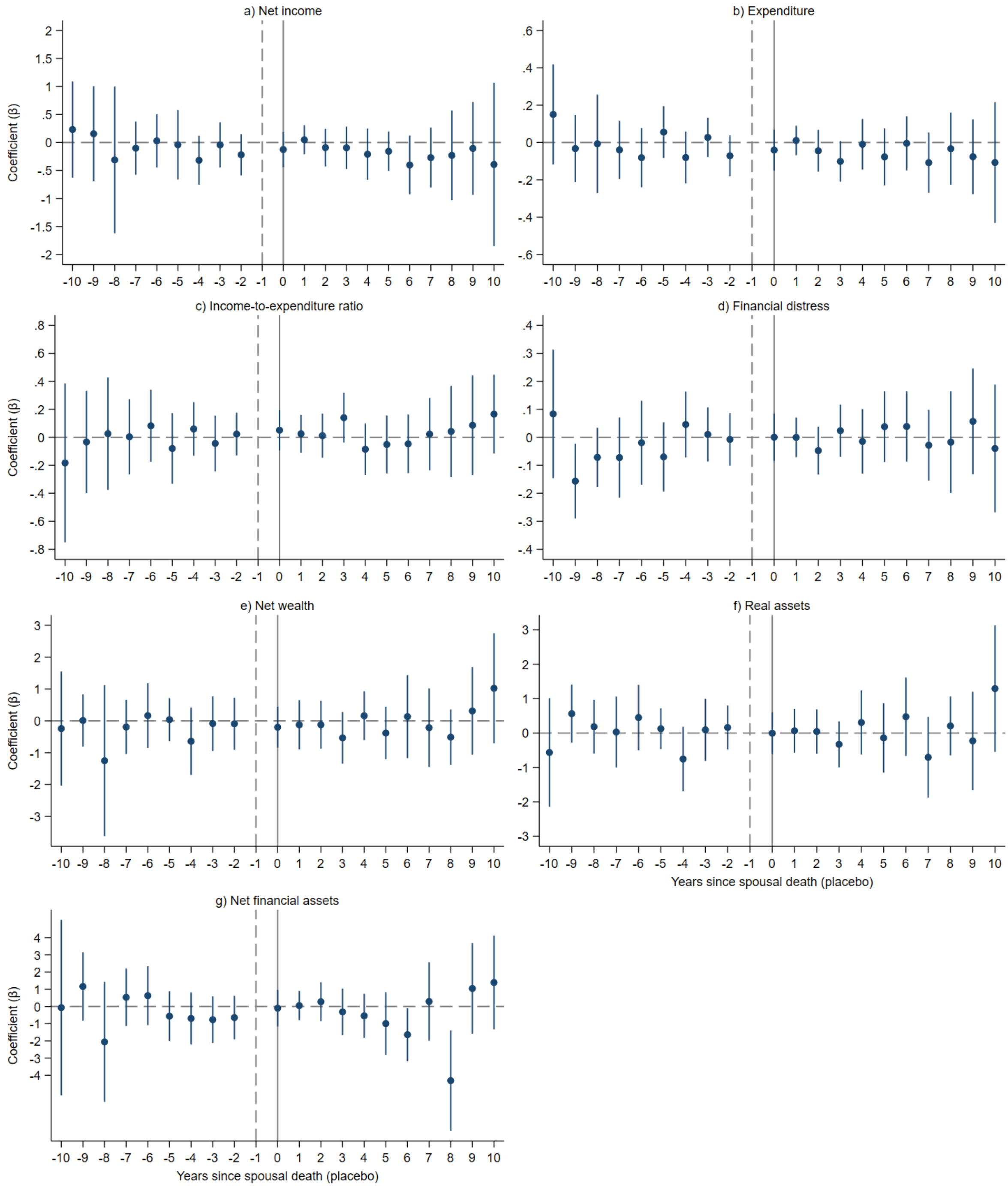
Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 12,328 and 12,508. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.16. DiD estimates on main outcomes, financial respondent status switchers

	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Net wealth: Real assets	Net wealth: Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.296***	0.0117	-0.226***	0.0498*	-0.198	-0.614***	-0.372
	(0.0817)	(0.0287)	(0.0394)	(0.0273)	(0.194)	(0.229)	(0.281)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimation sample varies across imputations; sample sizes vary between 11,172 and 11,352. Number of groups and clusters varies among imputations. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Figure A.2. Placebo test



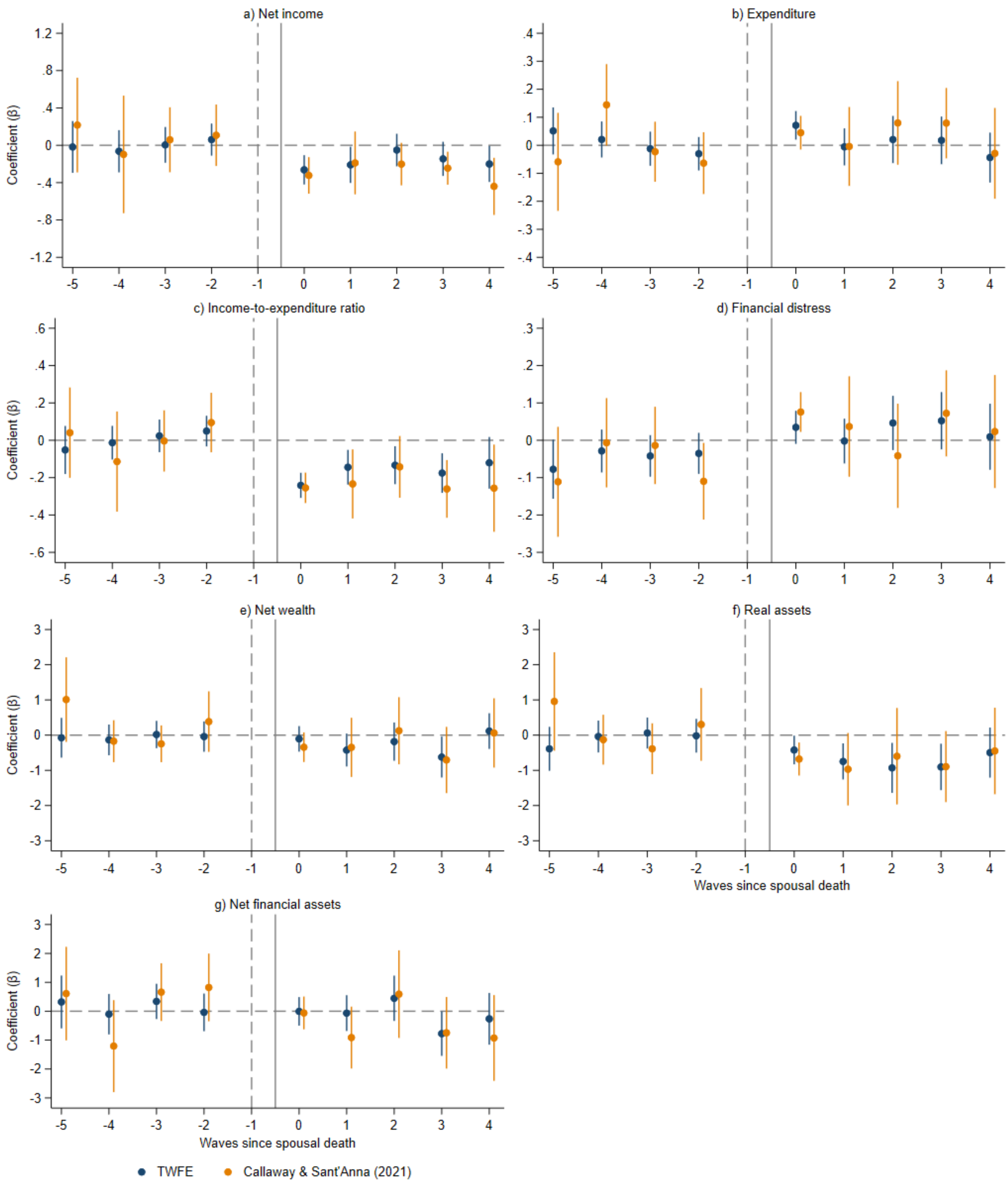
Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors are clustered at the individual level. 95% confidence intervals. Estimation sample varies across imputations; sample sizes vary between 11,902 and 12,082. Number of groups and clusters varies among imputations. $k = -1$ is the reference year (vertical dashed line), $k = 0$ is the year in which the placebo event happens (vertical solid line). All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.17. DiD estimates on main outcomes, not-yet treated as control group

	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Net wealth: Real assets	Net wealth: Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.179**	0.0851***	-0.230***	0.0268	-0.289	-0.580***	0.0934
	(0.0716)	(0.0288)	(0.0344)	(0.0231)	(0.197)	(0.221)	(0.267)
Individuals	1,119	1,119	1,119	1,119	1,119	1,119	1,119
Observations	4,476	4,476	4,476	4,476	4,476	4,476	4,476
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Results from five multiply imputed datasets. Coefficients and standard errors are pooled using Rubin's rule. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help*.

Figure A.3. Event study addressing dynamic treatment, dataset $m = 1$



Notes. Results from the first imputed dataset ($m = 1$). Standard errors are clustered at the individual level. 95% confidence intervals. $k = -1$ is the reference wave (vertical dashed line), $k = 0$ is the first wave after the event, which happens in between (vertical solid line). All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

Table A.18. DiD estimates on main outcomes, no imputations

						Net wealth	
	Net income	Expenditure	Income-to-expenditure ratio	Financial distress	Net wealth	Real assets	Net financial assets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DiD	-0.126	0.0288	-0.157**	0.0369	-0.862**	-0.888***	-0.515
	(0.160)	(0.0299)	(0.0624)	(0.0227)	(0.382)	(0.255)	(0.372)
Individuals	866	1,007	787	1,071	801	954	936
Observations	1,828	2,794	1,430	3,786	1,708	2,531	2,410
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All outcomes, except financial distress (binary), are PPP-adjusted, equivalised, and IHS-transformed. Controls include: *age categories, current job status, self-rated health, BMI categories, number of chronic diseases, number of mobility limitations, number of ADL limitations, number of IADL limitations, EURO-D, self-rated eyesight, self-rated hearing, times seen/talked to medical doctor, hospitalised, ever smoked daily, physical inactivity, self-rated reading skills, self-rated writing skills, orientation score, verbal fluency score, numeracy score, area of residence, number of rooms, moved, number of children, number of grandchildren, co-living, received help.*

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SHARE acknowledgements and data citation

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